

ERIKA BUSSMANN

erika-bussmann.github.io | linkedin.com/in/erika-bussmann | erikabussmann@google.com

SUMMARY

Machine Learning Engineer at Google DeepMind leading model quality, evaluation, and tool-use for Gemini Live in Chrome. Core research contributor to GDM Responsibility project using conversational geometry to study emergent dynamics in Human-AI interaction.

INDUSTRY EXPERIENCE

Research Scientist – Responsible AI, Google DeepMind

- Led development of core ground-truth datasets of interaction dynamics across 21K+ user-model conversations; scaled the dataset via a suite of auto-evaluations using LLM-as-a-Judge and CoT analysis, reducing false positive annotations by 5 to 28% across labels.
- Designed and iterated on per-dynamic human evaluation rubrics; analyzed rater agreement for per-turn and conversation-level annotations, and aligned auto-eval behavior with human ratings via iterative rubric refinement and few-shot calibration.
- Identified top interpretable embedding-based features driving interaction dynamics; established robust baselines for predictive power across feature sets (raw embeddings, language-style matching, Jaccard similarity) and techniques (random forest, CNNs).
- Designed and implemented experiments over featurized embedding time-series data to identify “signatures” of interaction dynamics using clustering techniques across multiple time horizons (per-turn, multi-turn window, conversation, multi-session).

ML Engineer – Gemini App, Google DeepMind

- Developed an LLM-based data flywheel to filter and curate high-quality eval sets representing the distribution of live Gemini traffic; adopted across all Chrome modeling workstreams and responsible for 30% of all quality regression and hill-climbing eval sets.
- Led refresh of the Gemini in Chrome eval set used to gate model releases on quality regressions, replacing a manual process that saved O(weeks) of eng. time, increased refresh frequency from bi-annual to on-demand, and reduced eval set flakiness by 8%.
- Developed an LLM-based pipeline to improve precision and recall of tool triggering, including unsupervised clustering of triggering conditions, classification of legitimate tool calls and failure modes, interpretability via CoT analysis, and fine-tuning via APO; reduced Google Search trigger rate by 18% without statistically significant loss in quality.
- Led modeling and evals for Gemini Live in Chrome — designing eval sets, implementing metrics, tuning LLM-as-judge, and building the eval harness for product CUIs and browser-context triggering; designed and executed end-to-end human evals for major launches.
- Built the Gemini in Chrome PoC for evaluating new Live model release candidates; upstreamed critical components to the A2A team.

ML Engineer – Auction Quality, Google Ad Exchange

- Optimized throttling algorithms and bid-likelihood regression models to maximize publisher revenue and buyer bidding efficiency; built simulations from historical bidding data to predict revenue headroom and designed live-traffic A/B experiments.
- Designed and launched multiple “experiments of experiments” to eliminate artificial discrepancies between throttling optimization control and holdback arms, reducing inaccuracy in experiment metrics by 5–50% across 45 bidders and 4 global endpoints.

Software Engineer – Obtainability & Efficiency, Google Compute Engine

- Designed and owned the primary data source of truth from prototype to GA launch for a patent-recognized compute obtainability optimization system, used by major GCP clients (e.g., Kubernetes) to maintain strict fault isolation compliance.

EDUCATION

Brown University, B.S., Computer Science & Applied Math — Providence, RI

PUBLICATIONS & PATENTS

- Cloud Obtainability Optimization and Stockout Experience — *U.S. Patent*
- Quantifying Resilience to Recurrent Ecosystem Disturbances Using Flow-Kick Dynamics — *Nature*
- Resilience of Socially Valued Properties of Natural Systems to Repeated Disturbance: A Framework to Support Value-Laden Management Decisions — *Natural Resource Modeling*
- Computer Aided Diagnosis of Skin Lesions from Morphological Features — *International Skin Imaging Collaboration*

SKILLS & INTERESTS

LLMs: Post-training evals (querysets, metrics, vendors, analysis), synthetic data generation, loss analysis / quality hillclimbing

Research Interests: Evaluation science, Human-Computer Interaction, Scalable Oversight

Languages / Tools: C++, Python, SQL, TensorFlow, Pandas